

1. Trophic Level?

Trophic Level • Trophic level of an organism is position it occupies in a food chain. (PPTS)

The trophic level of an organism is the position it occupies in a food web. A food chain is a succession of organisms that eat other organisms and may, in turn, be eaten themselves. The trophic level of an organism is the number of steps it is from the start of the chain. (Internet)

2. Ecological niche?

• Each organism in a community confronts the challenge of survival in a different way. • Occupies a specific functional role and place called its niche. • The niche an organism occupies is the total of all the ways it uses the resources of its environment.

A niche may be described in terms of: • Space utilization • Food consumption • Temperature range • Appropriate conditions for mating • Requirements for moisture • Other factors. (PPTS)

3. Trichocyst?

Trichocysts are pellicular structures primarily used for protection. • They are rodlike or oval organelles • Oriented perpendicular to the plasma membrane • Pellicle can discharge trichocysts, which then remain connected to the body by a sticky thread.

4. Adaptedness or Fitness? Explain Adaptedness?

Adaptedness or fitness: is a measure of the capacity for successful reproduction in a given environment. • Not every characteristic is an adaptation to some kind of environmental situation. • Neither adaptations lead to perfection.

5. Source of land pollution?

Pollution Classified as: 1. Point source 2. Non point source 1. Point source Single identifiable source of pollution Example: • Land fill • Factory drain • Sewages pipes

Non Point source Resulting from many diffuse sources Example: • Sediment • Nutrients • Toxic contaminants and chemicals • Pathogens

6. Logistic population growth?

Logistic population growth occurs when the **growth** rate decreases as the **population** reaches carrying capacity. Carrying capacity is the maximum number of individuals in a **population** that the environment can support.

7. Types of Genetics drift?

There are two major **types** of **genetic drift**: population bottlenecks and the founder effect. A population bottleneck is when a population's size becomes very small very quickly. This is usually due to a catastrophic environmental event, hunting a **species** to near extinction, or habitat destruction.

8. How natural selection work in evolution?

The mechanism that Darwin proposed for **evolution** is **natural selection**. Because resources are limited in **nature**, organisms with heritable traits that favor survival and reproduction will tend to leave more offspring than their peers, causing the traits to increase in frequency over generations.

9. R selected organism?

• High growth rates • Typically exploit less crowded ecological habitats • Produce many offspring • Relatively low probability of surviving to adulthood • High reproductive rate supports their survival. • Especially in unstable environment.

10. Ozone depleting substance?

Ozone depleting substances

include: □ chlorofluorocarbons

(CFCs)

- hydrochlorofluorocarbons (HCFCs) □ hydrobromofluorocarbons (HBFCs) □ halons.
- methyl bromide.
- carbon tetrachloride. □ methyl chloroform.

11. Nematode characteristic?

- Their body is bilaterally symmetrical and triploblastic.
- They are cylindrical in shape.
- They exhibit tissue level organization.
- Their body has a cavity or pseudocoelom.
- The alimentary canal is distinct, with the mouth and the anus.
- They are sexually dimorphic. (**Internet**)

12. Food chain?

A food chain is a linear network of links in a food web starting from producer organisms and ending at apex predator species, detritivores, or decomposer species. A food chain also shows how the organisms are related with each other by the food they eat. Each level of a food chain represents a different trophic level.

13. Endomembrane system?

What is the **endomembrane system**? The **endomembrane system** (endo- = “within”) is a group of membranes and organelles in eukaryotic cells that works together to modify, package, and transport lipids and proteins.

14. Primary productivity of wetland?

In general, swamps and marshes have the highest **primary** production of all the world's ecosystems. **Primary** production of all **wetland** types varies from 600-2000 gC/m²/y. ... So **wetlands** that are stagnant are less **productive** than those that flow or are open to flooding rivers.

12. Tropical rainforest ecosystem?

The **tropical rainforest** biome has four main characteristics: very high annual rainfall, high average temperatures, nutrient-poor soil, and high levels of biodiversity (species richness). ... **Tropical rainforests** also have high humidity; about 88% during the wet season and approximately 77% in the dry season.

15. Bio magnification?

Biomagnification, also known as bioamplification or **biological magnification**, is any concentration of a toxin, such as pesticides, in the tissues of tolerant organisms at successively higher levels in a food chain.

16. Phyletic gradualism?

Evolutionary changes are occurring gradually over millions of years. • This concept, called phyletic gradualism.

17. Acid rain?

rainfall made so acidic by atmospheric pollution that it causes environmental harm, chiefly to forests and lakes. The main cause is the industrial burning of coal and other fossil fuels, the waste gases from which contain sulphur and nitrogen oxides which combine with atmospheric water to form acids.

18. How global warming contribute to destroy evolution?

Increasing **global** temperature means that **ecosystems** will change; some species are being forced out of their habitats (possibly to extinction) because of changing conditions, while others are

flourishing. Secondary **effects** of **global warming**, such as lessened snow cover, rising sea levels, and weather changes.

19. Which factor is limit to population size and growth?

Limiting factors regulate population size • Rate of Birth • Rate of Death/mortality • Dispersal • Food Resources • Space to grow • Environmental factors

20. Consumer resources ecosystem?

Consumer–resource interactions are the core motif of ecological food chains or food webs, and are an umbrella term for a variety of more specialized types of biological species interactions including prey-predator (see predation), host-parasite, plant-herbivore and victim-exploiter systems.

21. Characteristic of bacteria?

Bacteria are like eukaryotic cells in that they have cytoplasm, ribosomes, and a plasma membrane. **Features** that distinguish a **bacterial** cell from a eukaryotic cell include the circular DNA of the nucleoid, the lack of membranebound organelles, the cell wall of peptidoglycan, and flagella.

22. Speices and Soecution?

• Species is a group of populations in which genes are actually, or potentially, exchanged through interbreeding. • This definition has problems. • Taxonomists depend upon: 1. Morphological 2. Ecological information • Especially in case of fossils
Speciation • Formation of new species. • A requirement of speciation is that subpopulations are prevented from interbreeding. • This is called reproductive isolation. • No gene flow in case of reproductive isolation.

23. Way of reproductive isolation?

Ways of Reproductivelsolation

1. Premating isolation • Geographical Isolation - prevents mating from taking place • Barriers, such as rivers or mountain ranges, may separate subpopulations
2. Postmating isolation • Prevents successful fertilization and development • Even though mating may have occurred.

24. Energy in ecosystem?

The largest source of **energy** for an **ecosystem** is the sun. **Energy** that is not used in an **ecosystem** is eventually lost as heat. **Energy** and nutrients are passed around through the food chain, when one organism eats another organism. Any **energy** remaining in a dead organism is consumed by decomposers. ([Internet](#))

25. Cell of porifera?

Sponges have several **cell** types: * Choanocytes (also known as "collar **cells**"), flagellated **cells** which function as the sponge's digestive system, are remarkably similar to the protistan choanoflagellates. The collars are composed of microvilli and are used to filter particles out of the water. ([Internet](#))

26. Class of arthropoda?

There are 5 major **classes of arthropods**. Insects, centipedes, millipedes, arachnids and crustaceans. They all have their own characteristics to belong in that particular **class**. ([Internet](#))

27. Origin eukaryotic?

The **origin** of the **eukaryotic** cell is a milestone in the evolution of life, since **eukaryotes** include all complex cells and almost all multicellular organisms. The timing of this series of events is hard to determine; Knoll (2006) suggests they developed approximately 1.6–2.1 billion years ago. ([Internet](#))

28. Habitat destruction?

Habitat destruction occurs when natural **habitats** are no longer able to support the species present, resulting in the displacement or **destruction** of its biodiversity. Examples include harvesting fossil fuels, deforestation, dredging rivers, bottom trawling, urbanization, filling in wetlands and mowing fields. (Internet)

29. Adaption occur?

• Adaptation occurs when a change in a phenotype increases an animal's chance of successful reproduction. • Change in environment is the force behind adaptations. • It may result in the evolution of multiple new groups. • To exploit the changed environment

30. Carbon cycle?

The carbon cycle is the biogeochemical cycle by which carbon is exchanged among the biosphere, pedosphere, geosphere, hydrosphere, and atmosphere of the Earth. Carbon is the main component of biological compounds as well as a major component of many minerals such as limestone.

31. Subphylum mastigophora?

Mastigophora (phylum Protozoa, **subphylum** Sarcomastigophora) A superclass of protozoa, which employ one or more flagella for locomotion. There are two classes, and 19 orders. The superclass includes a wide variety of organisms, many of which are classified alternatively with the algae. (Internet)

32. Symbiosis?

Symbiosis Relationship between two different species living together

33. What is commensalism?

One species gain benefits while those of the other species is neither benefited nor harmed. Example: Remora fish and Sharks

34. Characteristic chordata?

Characteristics of Chordata. Animals in the phylum Chordata share four key features: a **notochord**, a **dorsal hollow nerve cord**, **pharyngeal slits**, and a **post-anal tail**.

35. Why soft fossil reject?

Soft-bodied organisms are much less likely to be preserved than those ... Thus the null hypothesis can be **rejected**,

36. Exosphere layer?

• It is located directly above the thermosphere. • Thin, atmosphere-like volume surrounding a planet • Molecules are gravitationally bound to that body • Density is too low for them to behave as a gas by colliding with each other.

37. Conventional aviation?

• The lowest layer • All weather conditions take place here. • Conventional aviation takes place here.

38. Allopatric speciation?

Allopatric speciation is **speciation** that happens when two populations of the same species become isolated from each other due to geographic changes. **Speciation** is a gradual process by which populations evolve into different species.

39. Autotroph, heterotroph and saprozoic?

Autotrophs • Synthesis of the organic compounds of their bodies from inorganic precursors such as: - CO₂ , H₂O,

NO₃ • using energy from an abiotic source - Light - inorganic oxidation reactions

Heterotrophs • Cannot synthesize organic compounds from inorganic precursors. • Eat organisms from a different population for energy • Liberating chemical-bond energy for metabolic use in

glycolysis, Krebs's cycle and electron transport chain Example: Animals **saprozoic**.. Obtaining nourishment by absorption of dissolved organic and inorganic materials, as certain protozoans and nematodes do. **(Internet)**

40. Ecosystem?

An ecosystem is a community of living organisms in conjunction with the nonliving components of their environment, interacting as a system. These biotic and abiotic components are linked together through nutrient cycles and energy flows. **(Internet)**

41. Three point thermosphere?

Properties • Temperatures will significantly drop if solar activity is limited. • Ions are far apart to transfer heat through conduction. • Around 160 kilometres, the density is so low that molecular interactions are too infrequent to permit the transmission of sound.

42. Component water pollution?

It includes detergents and other cleaning materials, surfactants, pesticides and other organic industrial wastes. Many of these materials are poisonous for living organisms and cause serious **water pollution** problems. Detergents and other cleaning materials form an important source of **water pollution**. **(Internet)**

43. Omnivore?

An **omnivore** is a kind of animal that eats either other animals or plants. Some **omnivores** will hunt and eat their food, like carnivores, eating herbivores and other **omnivores**. Some others are scavengers and will eat dead matter. Many will eat eggs from other animals. **(Internet)**

44. Rate of evolution?

Rate of evolution is a measurement of the **rate** of genotype change of species and organisms over a period of time. **(Internet)**

45. Class of plasmodium?

Aconoidasida **(Internet)**

46. Type of pseudopodia?

pseudopodia can be assigned to one of four **types**: filopodia, lobopodia, rhizopodia, and axopodia. Lobopodia (Fig. 3.8), the most common **form** among parasitic amoebae, are blunt and may be composed of both ectoplasm and endoplasm or of ectoplasm only. **(Internet)**

47. Idiobiont and Koindiont?

- **Idiobiont** parasitoids • Sting their prey on capture • Either killing them outright or paralysing them immediately.
- **Koinobiont** parasitoids • Flies as well as wasps • Lay their eggs inside young hosts, usually larvae.
 - Allowed hosts to grow • Parasitoids emerge as adults, leaving the prey dead, eaten from inside

48. Mesosphere?

The **mesosphere** (/ˈmɛsoʊsfɪər/; from Greek mesos, "middle") is the third layer of the atmosphere, directly above the stratosphere and directly below the thermosphere. In the **mesosphere**, temperature decreases as altitude increases.

49. Mitochondria composition?

A **mitochondrion** contains outer and inner membranes composed of phospholipid bilayers and proteins. The two membranes have different properties. Because of this double-membraned organization, there are five distinct parts to a **mitochondrion**.

Long 5 Marks

50. Soft inheritance of acquired characteristic?

Theory of “inheritance of acquired characteristics or soft inheritance” • Lamarck noticed that organisms adapted to a particular niche had well developed specialized organs • For example a carnivore will have long canine teeth to grip its prey. • Small non-functional organs (vestigial organs). • e.g. the appendix in humans, the internal hind limbs of whales and the internal legs of some species of snakes

51. Phylum ciliophora?

The ciliates are a group of protozoans characterized by the presence of hair-like organelles called cilia, which are identical in structure to eukaryotic flagella, but are in general shorter and present in much larger numbers, with a different undulating pattern than flagella.

52. Phylum Platyhelminthes?

Phylum Platyhelminthes. **Phylum Platyhelminthes** is composed of the **flatworms**: acoelomate organisms that include many free-living and parasitic forms. Most of the **flatworms** are classified in the superphylum Lophotrochozoa, which also includes the mollusks and annelids.

53. Type of parasites?

- **Protozoa** - one-celled organisms that live and multiply in the blood or tissue of humans. ...
- **Helminths** - parasitic flatworms, flukes, tapeworms, thorny-headed **worms**, roundworms, and pinworms. ...
- **Ectoparasites** - ticks, fleas, lice, and mites that live on the surface of a human host and attach or burrow into the skin.

54. Autotroph, cytoskeleton and effect of aposematism?

Autotroph

Autotrophs • Synthesis of the organic compounds of their bodies from inorganic precursors such as: - CO₂ , H₂O, NO₃ • using energy from an abiotic source - Light - inorganic oxidation reactions

55. Cytoskeleton

Cytoskeleton provides support and shape to cell • Present in both eukaryotic and prokaryotic cell. • It is a network of protein fibers • Supporting cell shape • Anchoring organelles within the cell

56. Effect of aposematism:

- Appearance of an animal that warns predators it is toxic, distasteful or dangerous. • Advertisement of signals 1. Color 2. Odours 3. Sound 4. Aggression

57. Trophic level in food chain?

A **trophic level** is the group of organisms within an ecosystem which occupy the same **level** in a **food chain**. ... The second **trophic level** consists of herbivores, these organisms gain energy by eating primary producers and are called primary consumers. **Trophic levels** three, four and five consist of carnivores and omnivores. **(Internet)**

58. Invasive species?

- Colonization, a natural process by which a species expands its geographic range, occurs in many ways. Examples: • Seed dispersal by birds • Lowering of sea levels join two isolated land masses • Flooding.

59. Environmental effect on population growth in term of logistic growth?

In **logistic growth**, **population** expansion decreases as resources become scarce. It levels off when the carrying capacity of the **environment** is reached, resulting in an S-shaped curve. **(Internet)**

60. Method of water treatment?

The **methods** used include physical processes such as **filtration**, sedimentation, and distillation; biological processes such as slow sand filters or biologically active carbon; chemical processes

such as flocculation and chlorination; and the use of electromagnetic radiation such as ultraviolet light. **(Internet)**

61. Land pollution source?

The **source** of **land pollution** comes from the human element such as littering, and waste that is washed ashore from boats, oil rigs, and sewage outlets. **Land pollution** is a result of dumping garbage, waste, and other toxins making the **land** contaminated or **polluted**. **(Internet)**

62. Gene flow can cause R constrain evolution changes?

- Gene flow can impede adaptation within a population by the continual flow of inferior alleles from other populations.
- Natural selection is removing inferior alleles while gene flow can bring them back. Example: • Heavy metals tolerance

63. How much carbon dioxide produced by gasoline?

- 20% and 13 % of carbon dioxide emissions comes from the burning of gasoline in internal combustion engines of cars and trucks

64. K and R selected species?

k-selected species • Living at densities close to carrying capacity • Typically are strong competitors in such crowded habitats. • Invest more heavily in fewer offspring • Relatively high probability of surviving to adulthood.

r-selected species • High growth rates • Typically exploit less crowded ecological habitats • Produce many offspring • Relatively low probability of surviving to adulthood

65. Define plasmodium and life cycle?

Plasmodium • Causes malaria in humans • Very common apicomplexan. • Globally, an estimated 3.4 billion people in 92 countries are at risk of being infected with malaria.

Plasmodium species involve development in a blood-feeding insect host which then injects parasites into a vertebrate host during a blood meal. Parasites grow within a vertebrate body tissue (often the liver) before entering the bloodstream to infect red blood cells. **(Internet)**

66. Camouflage and mimicry?

Camouflage • Use of any combination of materials, coloration, or illumination • Concealment from predators • Also known as cryptic coloration

Mimicry Disguising them as something else

67. Binomial nomenclature?

Binomial nomenclature, also called binominal nomenclature or binary nomenclature, is a formal system of naming species of living things by giving each a name composed of two parts, both of which use Latin grammatical forms, although they can be based on words from other languages.

Long 10 Marks

68. Different type of survivorship curves?

Survivorship Curve • Y-axis is a logarithmic plot of numbers of survivors • X-axis is a linear plot of age

Type 1 (Convex) • Populations survive to an old age, then die rapidly. • Unimportance of environmental factors in influencing mortality • Potential life span • Example: Humans

Type II (Diagonal) • Constant probability of death throughout lives • Influence of environmental factors on mortality • Mortality independent of age • Example: birds, rodents

Type III (Concave) • High juvenile mortality • Influence of environmental factors and less resistance of juveniles on mortality • Lower mortality in adults • Example: Fish

69. Mosaic evolution?

- A species is a mosaic of different molecules and structures that have evolved at different rates.
- Some molecules or structures are conserved in evolution. • Others change more rapidly. • The basic design of a bird provides a simple example.
- All birds are easily recognizable as they because of highly conserved structures 1. Feathers 2. Bills 3. Body form • Particular parts of birds, however, are less conservative and have a higher rate of change.
- Wings have been modified for: 1. Hovering (humming bird) 2. Soaring (eagle, vulture) 3. Swimming (swans, ducks) • Legs have been modified for: 1. Wading (stork, heron) 2. Swimming (penguin, duck) 3. Perching (sparrow, parrot)

70. Adaption of parasites?

To lead a **parasitic** mode of life, the **parasites** have **adapted** themselves in such a way as to survive and adjust itself with the body environment of their host. **Adaptation** is a dynamic process of adjustment with the new environment for establishment, self regulation, self preservation and race continuation. **(Internet)**

71. Metamerism and tagmatization?

Metamerism • Segmented arrangement of body parts in an animal is called Metamerism. • Each segment having its own excretory, nervous and circulatory system.

Tagmatization • Organization of segments into functional units (e.g. head, thorax and abdomen)

72. Climate changes forest genetic resources?

Climate change poses special challenges both to **forests** and trees and to the **forest genetic resources** that are necessary to adapt to **climate change**. Diversity of **forest genetic resources** provides the adaptive potential for a species (or a population) to cope with **climatic changes** and future challenges. **(Internet)**

73. Global Warming?

Global warming is the long-term rise in the average temperature of the Earth's climate system. It is a major aspect of climate change and has been demonstrated by direct temperature measurements and by measurements of various effects of the warming. Global warming and climate change are often used interchangeably. **(Internet)**

74. Mode of selection with example?

There many ways of dividing up and categorizing Natural **Selection**. For **example**, there are the trichotomies of Natural **Selection**, Sexual **Selection** and Artificial **Selection**, and **Modes of Selection** (Stabilizing, Directional, and Disruptive) trichotomy. **(Internet)**

75. Detail note on lysosome ?

Lysosomes work like the digestive system to break down, or digest, proteins, acids, carbohydrates, dead organelles, and other unwanted materials. They break up larger molecules into smaller molecules. Those smaller molecules can then be used again as building blocks for other large molecules. **(Internet)**